Iordan Iordanov

Research & Software Engineer — Computational Geometry, Applied ML, and Engineering Leadership

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14+ YEARS SOFTWARE DEVELOPMENT 6+ YEARS AI / MACHINE LEARNING 6+ YEARS ENGINEERING MANAGEMENT

PROFILE

Senior software engineer and computer science researcher with a PhD in computational geometry and strong experience in applied AI R&D. 14+ years across research, production engineering, and technical leadership. Contributor to CGAL, the largest open-source library for computational geometry. As CTO of an AI startup, scaled the engineering organization from 10 to 70, built an AIOps platform end-to-end, and delivered 50+ client-facing projects. Expert in C++ and Python. Fluent in five European languages. Currently based in Tokyo; relocating to Europe.

EXPERIENCE

Researcher · Braid Technologies Tokyo, Japan — Sep 2025 – Present

C++, Python, CGAL, VTK, libigl, Eigen, pybind11, CMake

- Lead R&D on geometry generation and augmentation for manufacturing: quality-driven mesh analysis and improvement pipelines and novel methods for procedural surface processing.
- Ran a comparative evaluation of remeshing approaches to establish the production meshing strategy.
- Shipped maintainable Python bindings for the core C++ geometry engine, opening it to new application domains, and modularized a monolithic codebase into independently buildable libraries.

Chief Scientist → CTO / Director · Corpy & Co., Inc. Tokyo, Japan — Jul 2019 – Jul 2025

Python, PyTorch, NumPy, Docker, Celery, MySQL, JavaScript, Linux

- Built a production AIOps platform from scratch (data management, annotation, augmentation, training, evaluation, deployment, monitoring) serving 8 model families across ~140 datasets at ~95% uptime; deployed to 10+ trial clients, 2 converted to paying production.
- Cut build-to-deployment from ~1 hour to under 5 minutes (≈12x faster releases) with a custom Docker pipeline; included edge deployment via model quantization and exposed training/inference through REST APIs and gRPC for client integration.
- Delivered 50+ client AI/ML projects end-to-end across manufacturing, automotive, healthcare, and industrial AI; owned the full lifecycle from scoping through staffing, QA, delivery, and support, with up to 5 in parallel.
- Won 4 competitive government-funded research collaborations with a Japanese national research institute on AI quality-assurance guidelines (autonomous-driving vision, anomaly detection, fairness in financial risk, LLM QA); built a model-agnostic, task-extensible LLM evaluation framework.
- Scaled the engineering organization from 10 to 70 people; designed and ran the company OKR system from scratch across 35+ engineers per cycle, owning the full hiring pipeline, performance management, and ISMS / ISO process standardization toward IPO readiness.

R&D Engineer · Gamestream Ludres, France — Jan 2019 – Jun 2019

C++, OpenGL

- Developed a low-latency game-streaming client for Samsung TVs (full-HD at 60fps) using a proprietary SDK; researched frame upscaling based on neural networks.

PhD Candidate · LORIA / Université de Lorraine Nancy, France — Jan 2016 – Dec 2018

C++, CGAL

- Formulated and implemented an algorithm for periodic Delaunay triangulations of the Bolza surface (the most symmetric compact hyperbolic surface of genus 2), now part of CGAL; published at SoCG 2017 and Curves & Surfaces 2018.

Graduate Research Assistant · IACM-FORTH Heraklion, Greece — Dec 2014 – Oct 2015

Python, C++, VMTK

- Applied geometric processing to medical research on abdominal aortic aneurysms; developed a geometry-based method for quantifying morphological change, published in Medical & Biological Engineering & Computing.

Software Developer · 01Sistemi Italy — May 2013 – Jun 2014

Java, InstantDeveloper

- Built business software, including an Italian cadastral PDF parser and a tax-calculation engine for the 730 tax declaration.

Software Developer · Nanosoft Greece — Dec 2009 – Mar 2012

VB.NET, WPF, Android

- Developed order-taking applications for Windows, Windows CE, and Android.

EDUCATION

PhD, Computer Science · Université de Lorraine Nancy, France 2016 – 2018
Thesis: Delaunay triangulations of a family of symmetric hyperbolic surfaces in practice

MSc, Applied Mathematics · University of Crete Heraklion, Greece 2014 – 2015
Thesis: Shape-Preserving Interpolation on the Sphere

BSc, Applied Mathematics · University of Crete Heraklion, Greece 2005 – 2013
Thesis: The Euclidean InSphere Predicate

SKILLS

Geometry & Math: Computational geometry · Delaunay triangulations · hyperbolic geometry · mesh processing · quad meshing / remeshing · NURBS · topology · CGAL · libigl · Eigen · VTK · PyVista · OpenCASCADE · FreeCAD · Blender

ML / AI: PyTorch · deep learning · computer vision · anomaly detection · object / pose / depth estimation · GANs · MLOps · model quantization · edge deployment · LLM evaluation · XAI · Quality assurance for AI (QA4AI)

Software Engineering: C++ · Python · pybind11 · CMake · Conan · UV · Docker · Celery · MySQL · Linux / Bash · full-stack web (JS) · Git · CI/CD · GCP / Azure / AWS (working familiarity)

Leadership: Org scaling (10→70) · OKR design and operation · full hiring pipeline · 50+ delivered projects · ISMS / ISO 27001 implementation

Languages: Native: Greek, Bulgarian · Fluent: English, Italian · Advanced: French · Basic: Japanese

PUBLICATIONS

5. Cache-Property-Aware Features for DNS Tunneling Detection
   N. Ishikura, D. Kondo, I. Iordanov, V. Vassiliades, H. Tode · ICIN, Paris, 2020
4. Systole of regular hyperbolic surfaces with an application to Delaunay triangulations
   M. Ebbens, I. Iordanov, M. Teillaud, G. Vegter · Curves & Surfaces, Arcachon, 2018
3. Delaunay triangulations of regular hyperbolic surfaces
   M. Ebbens, I. Iordanov, M. Teillaud, G. Vegter · Curves & Surfaces, Arcachon, 2018
2. Implementing Delaunay triangulations of the Bolza surface
   I. Iordanov, M. Teillaud · SoCG, pp. 44:1–44:15, 2017
1. A novel approach for local abdominal aortic aneurysm growth quantification
   E. Metaxa, I. Iordanov, E. Maravelakis, Y. Papaharilaou · Medical & Biological Engineering & Computing, 2016

OPEN SOURCE

CGAL — Computational Geometry Algorithms Library

- Contributed periodic Delaunay triangulations of the Bolza surface; announced February 2019 (cgal.org/2019/02/25/Hyperbolic_triangulations).